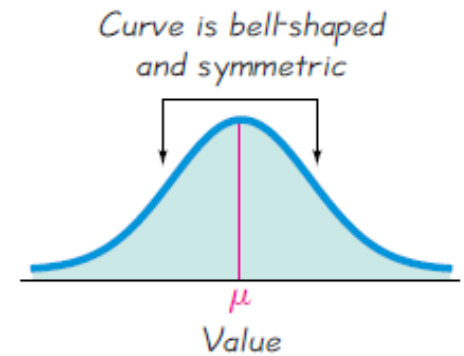


# Normal Distribution

A. prof. Elsayed Mahdy

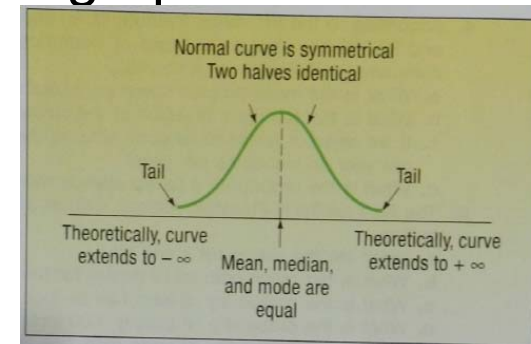
# 2- Normal Distribution

## “N.D.”



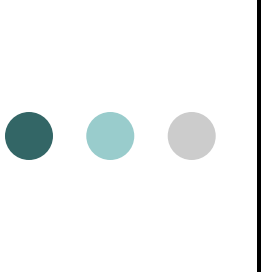
- **Definition:** A **continuous RV** that has a distribution with a **symmetric** and **bell-shaped** graph and described by the **PDF**

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\left[\frac{-(x-\mu)^2}{2\sigma^2}\right]}$$



is said to be **normally** distributed **with** **mean  $\mu$  and standard deviation  $\sigma$** .

- **Normal Distributions are extremely important?** because they occur so often and play important role in methods of **INFERENCE STATISTICS**.



# Normal Distribution Curve

(Back to MATH103) “if u have time”

- $f(x)$  has a relative and absolute maximum at  $\mu$ . It is Bell-shaped and it has a single peak at the center of the distribution. The mean, median and mode are equal and located at the center. Half the area under the normal curve is to the right of this center and the other half to the left of it.
- $f(x)$  is symmetric about the mean “vertical line  $x=\mu$ ”, i.e., the two halves will be mirror images. The tails of the curve extend indefinitely in both directions closer to the x-axis but never touches it.
- $f(x)$  has two inflection points at  $\mu \pm \sigma$ . The location of a normal distribution is determined by the mean  $\mu$ . The dispersion of the distribution is determined by the standard deviation  $\sigma$ .

# Try this at home! (for N.D.)

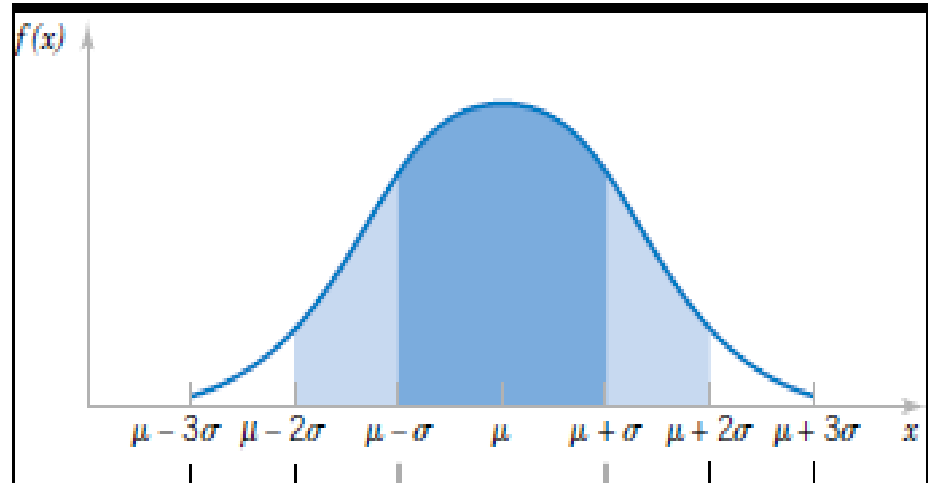
- Verify that

$$1 - \int_{-\infty}^{\infty} f(x) dx = 1.$$

$$2 - E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx = \mu.$$

$$3 - V(X) = \int_{-\infty}^{\infty} x^2 \cdot f(x) dx - \mu^2 = \sigma^2.$$

(Hint:  $z = \frac{x - \mu}{\sigma}$  simplifies the expression.)



# Computing Probabilities

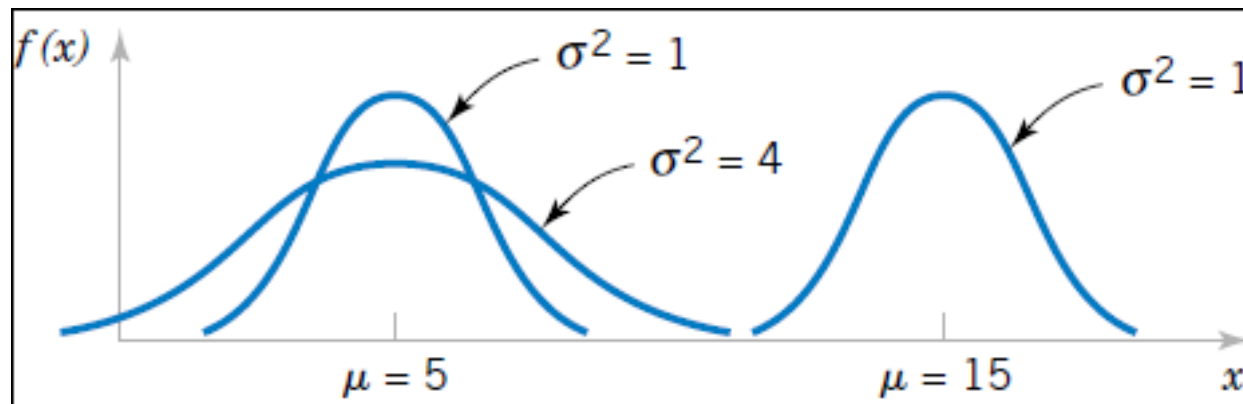
- To compute probabilities for a normally-distributed variable, we have to integrate

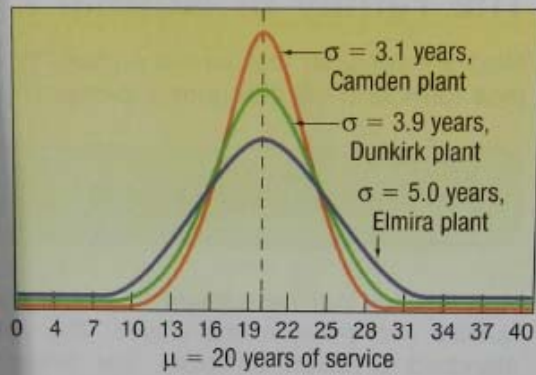
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$$

- Unfortunately, we cannot find an anti-derivative.
  - Will have to use statistics tables.
  - Tables are available only for the so-called standard normal distribution (S.N.D.).

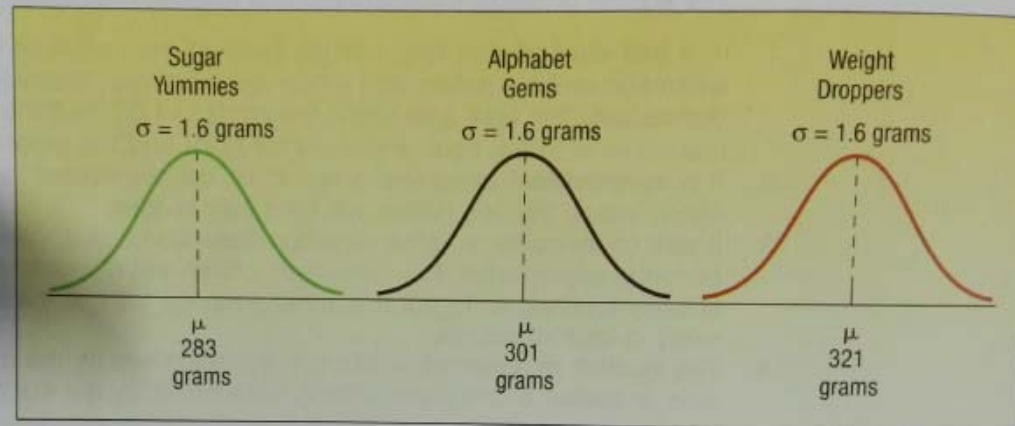
Which values of  $\mu$  and  $\sigma$  you would select if you work with N.D.?

- Since each normally distributed variable has its own  $\mu$  and  $\sigma$ , therefore the shape and the location of these curves will vary, i.e., there are so many different normal distributions, and you would have to have a table of areas under the curve for each variable!!!!!!!!!!!!!!
- Think of how the curve changes if one varies the parameters  $\mu$  &  $\sigma$ .

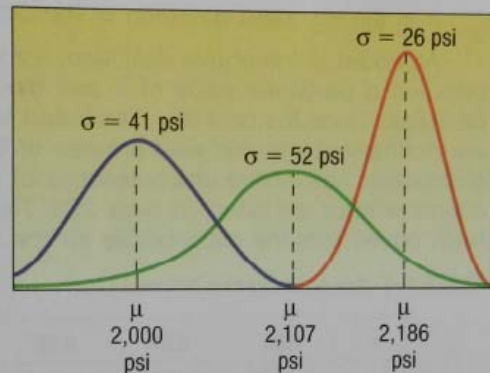




**CHART 7-4** Normal Probability Distributions with Equal Means but Different Standard Deviations



**CHART 7-5** Normal Probability Distributions Having Different Means but Equal Standard Deviations



**CHART 7-6** Normal Probability Distributions with Different Means and Standard Deviations

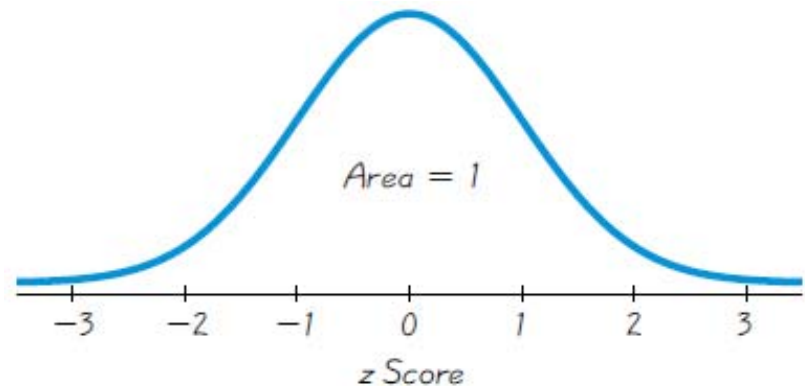
# 3- The Standard Normal Distribution “S.N.D.” “*z Dist.*”

- **Definition:** A normally distributed RV is said to have **standard N.D** if  $\mu = 0$  and  $\sigma = 1$ .

- It is traditionally denoted by **Z** and its PDF is

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{\frac{-z^2}{2}}$$

- The curve of S.N.D. is **bell-shaped**, **symmetric** about **z=0**.





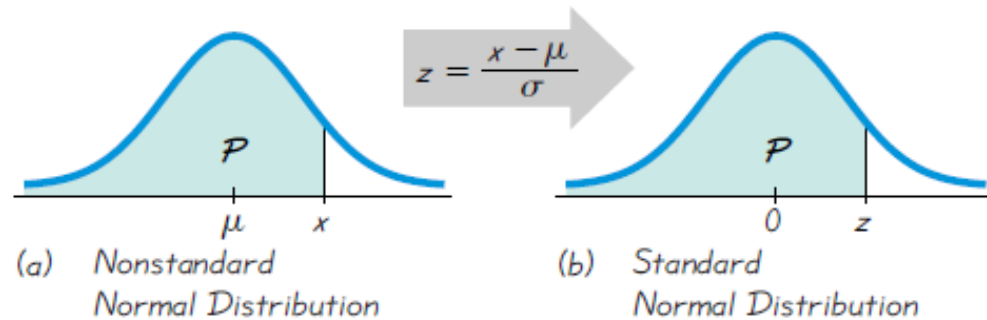
# CPF of Standard N.D.

- We only need to find a way to compute S.N.D. probabilities.
- It is known that it is impossible to express an anti-derivative  $\Phi$  of  $\phi$  explicitly:

$$\Phi(z) = P(Z < z) = \int_{-\infty}^z \phi(s) ds.$$

- However, there is a table with values of  $\Phi$ .
- Table for S.N.D. gives the area under the standard normal curve between “ $-\infty$ ” and any  $z$ .

# Convert any N.D. 2 S.N.D



- By definition,  

$$P(a \leq X \leq b) = \int_a^b f(x) dx = \frac{1}{\sigma \sqrt{2\pi}} \int_a^b e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx.$$

- Use the substitution

$$z = \frac{x - \mu}{\sigma} \Rightarrow dz = \frac{dx}{\sigma}$$

$$a \rightarrow \frac{a - \mu}{\sigma} = z_a \quad \& \quad b \rightarrow \frac{b - \mu}{\sigma} = z_b$$

- We get:

$$P(a \leq X \leq b) = \frac{1}{\sqrt{2\pi}} \int_{z_a}^{z_b} e^{-\frac{z^2}{2}} dz = \int_{z_a}^{z_b} \phi(z) dz = P(z_a \leq Z \leq z_b).$$

# Conclusion

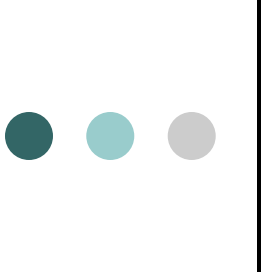
- A probability for a [non-standard] N.D. can be computed **via S.N.D.**

$$P(a \leq X \leq b) = P(z_a \leq Z \leq z_b),$$

$$\text{where } \frac{a - \mu}{\sigma} = z_a, \frac{b - \mu}{\sigma} = z_b.$$

- Hence,

$$P(a \leq X \leq b) = P(z_a \leq Z \leq z_b) = \Phi(z_b) - \Phi(z_a).$$



# STEPS: Convert any N.D. S.N.D.

1. Sketch a normal curve, label the mean and the specific  $x$  values, then shade the region representing the desired probability.
2. For each relevant value  $x$  that is a boundary for the shaded region, convert it to the equivalent  $z$  value.
3. Use the S.N.D. table to find the area of the shaded region. This area is the desired probability.

# How to use the z-Table.

**TABLE A-2** (continued) Cumulative Area from the LEFT

| z   | .00   | .01   | .02   | .03   | .04   | .05   | .06   | .07   | .08   | .09   |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 0.0 | .5000 | .5040 | .5080 | .5120 | .5160 | .5199 | .5239 | .5279 | .5319 | .5359 |
| 0.1 | .5398 | .5438 | .5478 | .5517 | .5557 | .5596 | .5636 | .5675 | .5714 | .5753 |
| 0.2 | .5793 | .5832 | .5871 | .5910 | .5948 | .5987 | .6026 | .6064 | .6103 | .6141 |
|     |       |       |       |       |       |       |       |       |       |       |
| 1.0 | .8413 | .8438 | .8461 | .8485 | .8508 | .8531 | .8554 | .8577 | .8599 | .8621 |
| 1.1 | .8643 | .8665 | .8686 | .8708 | .8729 | .8749 | .8770 | .8790 | .8810 | .8830 |
| 1.2 | .8849 | .8869 | .8888 | .8907 | .8925 | .8944 | .8962 | .8980 | .8997 | .9015 |
| 1.3 | .9032 | .9049 | .9066 | .9082 | .9099 | .9115 | .9131 | .9147 | .9162 | .9177 |
| 1.4 | .9192 | .9207 | .9222 | .9236 | .9251 | .9265 | .9279 | .9292 | .9306 | .9319 |

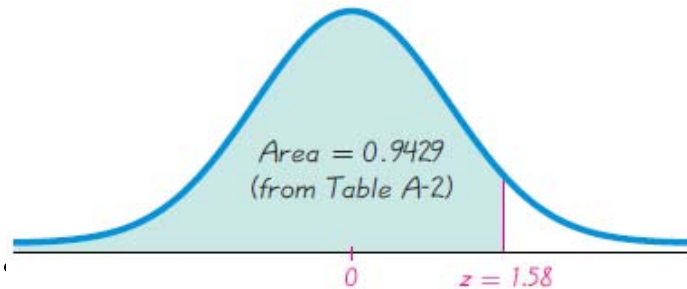
$$1 - \Phi(1.27) = P(Z \leq 1.27) = 0.8980$$

$$2 - \text{If } \Phi(z) = 0.8980 \Rightarrow z = 1.27$$

# Example 1

- Find the area under the S.N.D. curve between  $z = 0$  and  $z = 1.58$ .

- Solution**: Using the z-Table,  
The area between “ $-\infty$  & 1.58” =  
 **$P(Z < 1.58) = \Phi(1.58) = 0.9429$** .



- Therefore, the area between “0 & 1.58” is  
 **$P(0 < Z < 1.58) = P(Z < 1.58) - 0.5$**   
 **$= \Phi(1.58) - 0.5 = 0.9429 - 0.5 = 0.4429$** .

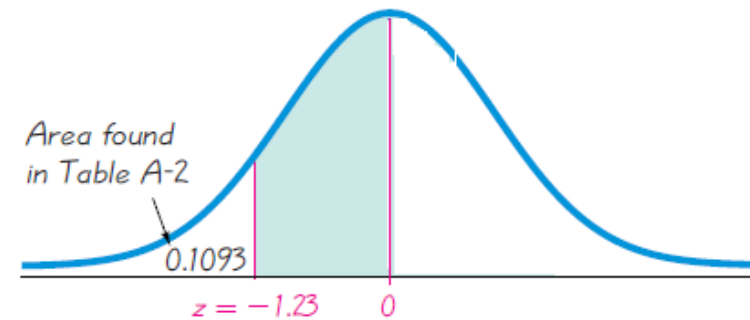
## Example 2

- Find the area under the S.N.D. curve between  $z = -1.23$  and  $z = 0$ .

- Solution**: Using the z-Table,

The area between “ $-\infty$  &  $-1.23$ ” =  
 $=P(Z < -1.23) = \Phi(-1.23) = 0.1093$ .

- Then, the area between “ $-1.23$  &  $0$ ” is  
 $= P(-1.23 < Z < 0) = 0.5 - P(Z < -1.23)$   
 $= 0.5 - \Phi(-1.23) = 0.5 - 0.1093 = 0.3907$ .



# Example 3

- Find the area under the S.N.D. curve between  $z = -2.00$  and  $z = 1.5$ .

- Solution**: Using the z-Table,

The area between “-2 & 1.5” is

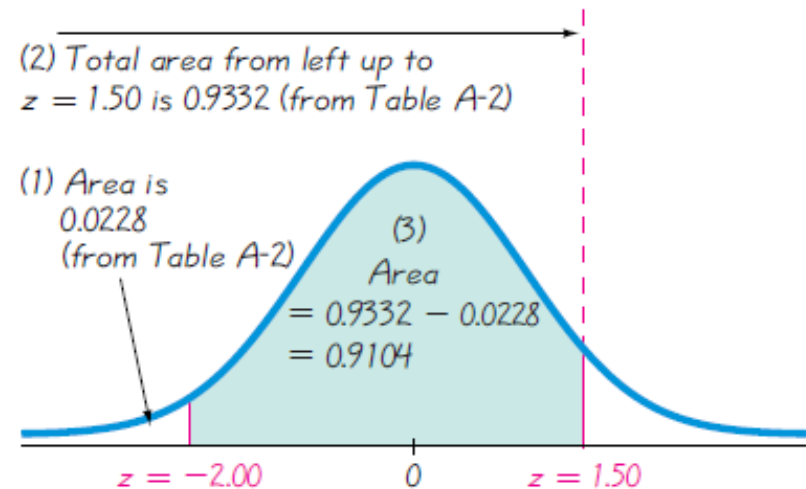
$$= P(-2 < Z < 1.5)$$

$$= P(Z < 1.5) - P(Z < -2.0)$$

$$= \Phi(1.5) - \Phi(-2.0)$$

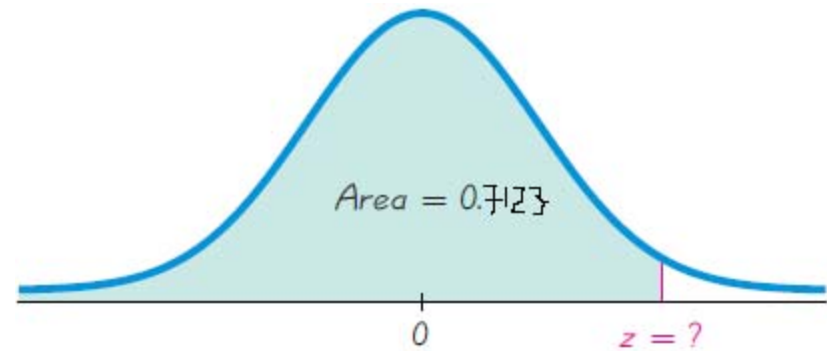
$$= 0.9332 - 0.0228$$

$$= 0.9104.$$



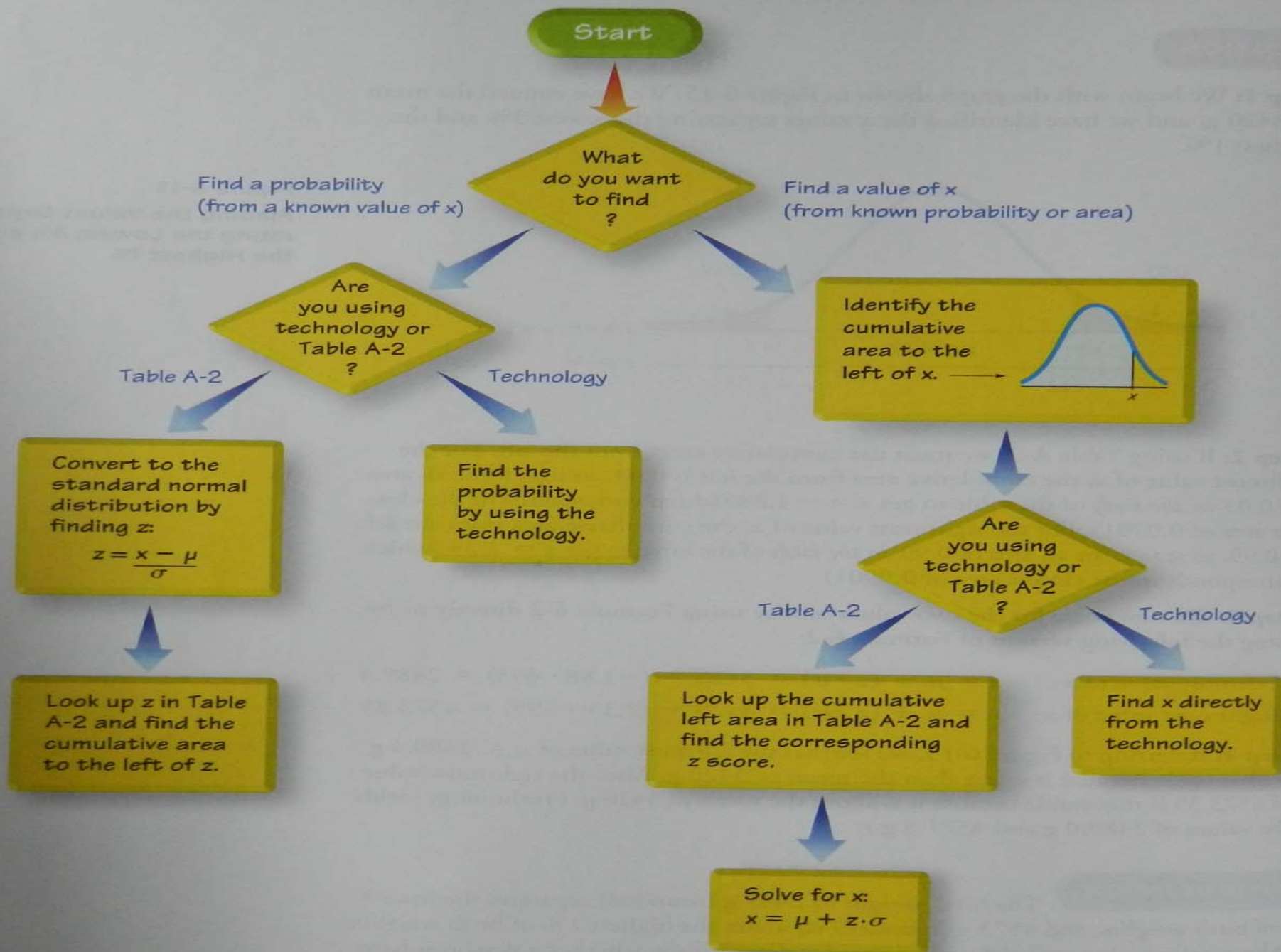


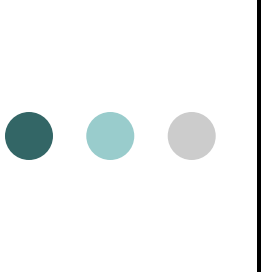
## Example 4



- Find the  $z$  value such that the area under the S.N.D. curve between 0 and the  $z$  value is 0.2123.
  - Solution**: Since  $P(0 < Z < z) = 0.2123$ .
  - Then, the area under the S.N.D. curve between " $-\infty$ " and the  $z$  value is:  
 $= P(Z < z) = \Phi(z) = 0.5 + 0.2123 = 0.7123$ .
  - Using the [z-Table](#), the area 0.7123 corresponds to  $z = 0.56$ .

# Applications with Normal Distributions





## Example 5

- Each month, an American household generates an **average** of **143** pounds of newspaper for garbage or recycling. Assume that **standard deviation** is **29** pounds. If a household is selected at random,  
find the probability of its generating
  - a) Between 140 and 211 pounds per month.
  - b) More than 174.9 pounds per month**Assume the variable is normally distributed.**

# Solution (a)

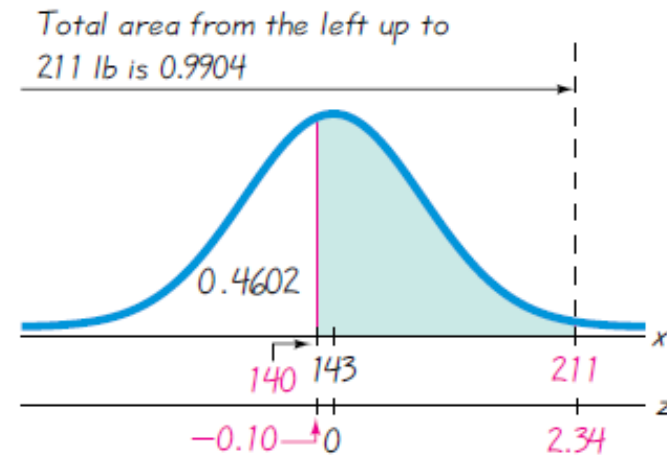
1. Find the two Z values corresponding to the given two X values:

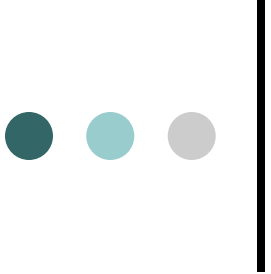
$$Z_1 = \frac{X_1 - \mu}{\sigma} = \frac{140 - 143}{29} = -0.1$$

$$Z_2 = \frac{X_2 - \mu}{\sigma} = \frac{211 - 143}{29} = 2.34$$

2. Hence,

$$\begin{aligned} P(140 < X < 211) &= P(-0.1 < Z < 2.34) \\ &= \Phi(2.34) - \Phi(-0.1) \\ &= 0.9904 - 0.4602 = 0.5302 \end{aligned}$$





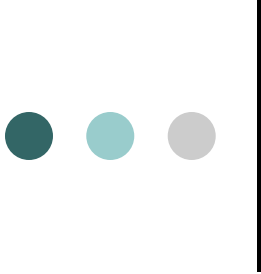
## Solution (b)

1. Find the appropriate  $z$  value.

$$z = \frac{X - \mu}{\sigma} = \frac{174.9 - 143}{29} = 1.1$$

2. Thus,

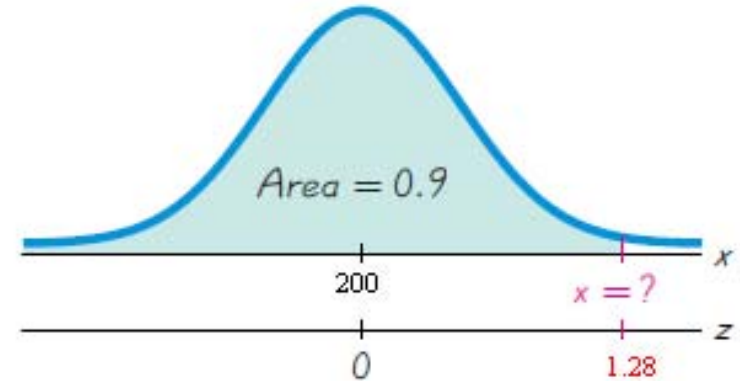
$$\begin{aligned} P(X > 174.9) &= P(Z > 1.1) \\ &= 1 - P(Z < 1.1) \\ &= 1 - \Phi(1.1) \\ &= 1 - 0.8643 = 0.1357. \end{aligned}$$



## Example 6

- **To qualify** for a police academy, candidates must score in the top **10%** on a general ability test. The test has a **mean** of **200** and a **standard deviation** of **20**. **Find the lowest possible score to qualify**. Assume the test scores are normally distributed.

# Solution



1. Need to find a value  $x$  such that the area from  $x$  to  $\infty$  is 0.1 (10%).
2. The area from “ $-\infty$ ” to  $x$  is  $1 - 0.1 = 0.9$ .
3. Use the table to get the corresponding  $z$  *to the area 0.9: since 0.9 is not an available area in the table, we use the closest area to 0.9 which is 0.8997, so the corresponding  $z$  value:  $z = 1.28$*
4. Get  $x = z \cdot \sigma + \mu = (1.28)(20) + 200$
5.  $x = 225.6$

THANK  
*you*